MIL-B-29186(YD)
27 October 1978

MILITARY SPECIFICATION

BARRIER SYSTEM: FLOATING, OIL AND DEBRIS CONTAINMENT (FOR PERMANENT INSTALLATION)

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a floating containment subsystem designed to be permanently installed for use in containing spilled oil and debris in harbors.

1.2 Classification of hardware. Hardware shall be one of the following types as specified (see 6.2.1):

(1) Boom connectors (pair).
(2) Boom connector adapter.
(3) Boom bulkhead assembly.
(4) Boom towing assembly.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-200 - Aluminum Alloy, Bar, Rod, Shapes, Tube, Wire Extruded, and Structural Shapes; General Specification for.
QQ-A-200/8 - Aluminum Alloy Bar, Rod, Shapes, Tube and Wire, Extruded, Structural Shapes.
PPP-B-601 - Boxes, Wood, Cleated Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1945

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MIL-P-116 - Preservation-Packaging, Methods of.
MIL-F-859 - Fuel Oil, Burner.
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-R-24049 - Rope, Polypropylene.

STANDARDS

FEDERAL

FED STD No. 595 - Color.
FED TEST METHOD STD No. 191 - Textile Test Methods.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

CIVIL ENGINEER SUPPORT OFFICE (CESO)

73-11-1F - Boom Connector.
73-11-2F - Boom Connector Adapter.
73-11-3F - Boom Bulkhead Assembly.
73-11-4F - Boom Tow Assembly.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from theprocuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A153 - Zinc Coating (Hot Dip) on Iron and Steel Hardware.
D975 - Specification for Diesel Fuel Oils.
D1141 - Substitute Ocean Water, Specification for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)

(Submission of requests for using biodegradeable materials in operational tests of this specification should be made by the contractor to the particular Regional Environmental Protection Agency Office in charge of the area.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The barrier shall consist of a single type of boom fitted with standard Navy boom hardware. The boom will be installed at pierside and in open harbor water. The boom shall be highly resistant to impact forces caused by docking vessels and shall be used in low wind and low current areas immediately adjacent to the pier. The boom shall also have a high static righting moment (see 3.6.3) for use in open harbor waters near the pier where strong current and high winds are likely to occur. In addition, the barrier shall have a repair kit with materials to make field repairs to the required booms. Figure 1 illustrates the use of the boom in the barrier system.

3.1.1 Hardware. Hardware shall be as specified on the applicable drawings forming a part of this specification. It shall be noted that the drawings are for Navy designed hardware and do not preclude the use of manufacturer's boom accessories, such as reinforcing and ballast, that might be termed hardware, providing such hardware is integral with the boom and compatible with Navy designed equipment without modification to the Navy equipment. Compatibility may not be accomplished by use of shackles, pins, dowels, or other loose hardware.

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3.1.2 Drawings. The drawings forming a part of this specification are engineering design drawings. The contractor is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the figures to insure correct fit, assembly, and operation of the item. No deviation from the prescribed dimensions or tolerance is permissible without prior approval of the contracting office.

3.2 First article. When specified (see 6.2.1), the contractor shall furnish samples and components of the barrier as specified in Section 4, for first article inspection and approval (see 4.2.1).

3.2.1 Hardware. When specified (see 6.2.1), the contractor shall furnish one boom connector and one boom connector adapter, one boom bulkhead assembly, and one boom tow assembly for first article inspection and approval (see 4.2.1 and 6.4).

3.2.2 First article test plan. When a first article test is required, the contractor shall submit a first article test plan for approval prior to conducting the first article tests (see 4.6.1 and 6.2.1).

3.3 Interchangeability. All units of the same type shall be functionally and dimensionally interchangeable. This requirement includes parts, assemblies, and accessories.

3.4 Materials. All materials used for boom construction shall be inherently resistant to, or be treated to become resistant to, the effects of continuous exposure to the following working environment: sea and brackish water; fresh water; weathering, both ultraviolet and ozone; and petroleum products (crude oil through refined and treated oil products). Ballast material shall be as specified in 3.5.6. Boom material, after use for oil containment, shall be easily cleaned with commercially available industrial detergents and water without damage to the materials. Materials used shall be free from defects which would adversely affect the performance of maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specifically stated.

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3.4.1 Hardware.

3.4.1.1 Aluminum. All aluminum shall be 6061-T6. The contractor shall provide a certificate of compliance to insure that all extruded aluminum meets the requirements of QQ-A-200 and QQ-A-200/8. All aluminum required in this specification shall be anodized as specified in MIL-A-8625, type II, class 1.

3.4.1.2 Stainless steel. All stainless steel shall be of standard quality suited for its intended use.

3.4.1.3 Galvanized products. All galvanized products shall conform to ASTM A153.

3.4.1.4 Rope. The polypropylene rope shall be as specified in MIL-R-24049, type I.

3.4.1.5 Shell. The flotation shell of the boom tow assembly shall be medium density polyethylene.

3.4.1.5.1 Foam. The filling required in the flotation shell shall be polyurethane foam.

3.5 Design.

3.5.1 Barrier components. The components of the containment system are shown schematically in Figure 2. The design of the boom shall incorporate the latest state-of -the-art developments for this type of equipment.

3.5.2 Boom dimensions and weights.

3.5.2.1 Freeboard. The height of the assembled boom above the free floating water line shall be no less than 12 inches nor more than 50 percent of the boom overall height. This freeboard height shall be constant the entire length of the boom with no bending or foldover of the boom material between floats and shall be maintained under all conditions of current from slack water to a current of 3 knots as specified in 4.6.2.2.3. If vertical stiffeners are used, they shall either be hermetically sealed within a fabric which is the same as the boom fabric and hermetically sealed to the boom fabric, or be otherwise designed to preclude snagging.

3.5.2.2 Draft. The draft of the barrier shall be of such a depth to insure a vertical attitude, as specified in 3.6.3, and an overall boom height of 36 inches is maintained.

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3.5.2.3 Flotation. Flotation shall be of a nonrigid closed cell solid foam type to insure that, in the event of a puncture or tear in its protective coating, there is no loss of freeboard below that of 3.5.2.1, or loss of flotation material to the environment. The floats may be either an integral or detachable part of the boom assembly. If floats are not integral to the boom assembly, field removal and replacement of damaged floats must be possible (see 3.6.5). Sufficient flotation shall be provided to result in a total buoyant force per foot no less than three times the total air weight per foot of the assembled boom, including end connectors. The flotation shall be completely enclosed by a coating meeting the requirements of 3.5.4.

3.5.2.4 Boom section length. A boom section shall be 50 feet in length and shall include a Navy standard male connector on one end, a female connector on the other end, anchor attachment points as specified in 3.5.9.1.1, and at least 10 above-water handholds as specified in 3.5.5.

3.5.2.5 Assembled weight. Each assembled 50 foot boom section, complete with end connectors and anchor points, shall weigh no more than 600 pounds (lb) in air. The requirements of this paragraph shall be demonstrated in the manner prescribed in 4.6.2.1.11.

3.5.3 Barrier material. The material forming the boom freeboard and draft shall meet all of the following requirements and shall be of sufficient stiffness to maintain the required freeboard along the length of the boom between floats or vertical stiffeners. Only solid sheet material or those woven fabrics having an impregnated or fully coated covering will be allowed. Fabrics with laminated coverings, which tend to delaminate upon continuous immersion, will not be allowed.

3.5.3.1 Tensile (breaking) strength. When tested in accordance with 4.6.2.1.1, the boom material shall exhibit a tensile strength no less than 500 lb in either the warp or fill direction.

3.5.3.2 Tear strength. When tested in accordance with 4.6.2.1.2, the boom material shall have a tear strength no less than 250 lb.

3.5.3.3 Puncture resistance. When tested in accordance with 4.6.2.1.3, the boom material shall have a puncture strength of no less than 150 lb.

3.5.3.4 Abrasion resistance. After being subjected to the abrasion condition specified in 4.6.2.1.4, the boom material shall exhibit a tensile strength of no less than 400 lb when tested in accordance with 4.6.2.1.1, a tear strength of 200 lb when tested in accordance with 4.6.2.1.2, and a puncture strength of no less than 125 lb when tested in accordance with 4.6.2.1.3.

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3.5.3.5 Weathering resistance. The boom material shall be subjected to the weathering test environment of 4.6.2.1.5. The boom material shall exhibit a tensile strength no less than 400 lb when tested in accordance with 4.6.2.1.1, a puncture strength of no less than 125 lb when tested in accordance with 4.6.2.1.3, and a tear strength of no less than 200 lb when tested in accordance with 4.6.2.1.2. When weathering and abrasion are combined, the boom material shall exhibit a tensile strength no less than 375 lb when tested in accordance with 4.6.2.1.1, a tear strength of no less than 190 lb, when tested in accordance with 4.6.2.1.2, and a puncture strength of no less than 100 lb when tested in accordance with 4.6.2.1.3.

3.5.3.6 Seam strength. Where material seams are present between boom section ends, they shall develop the full tensile strength of the parent material, as specified in 3.5.3.1 and as tested in 4.6.2.1.9. The boom material shall be reinforced at the location of attachment of the standard Navy end connectors. The connectors shall be joined to the boom material in such a manner as to minimize abrasion and tearing of the fabric under tensile load, as specified in 3.5.8.

3.5.3.7 Water absorption. The boom material properties of 3.5.3.1 through 3.5.3.3 shall not be degraded after continuous immersion in seawater as specified in 4.6.2.1.6. Further, when a coated fabric is used as the boom material, the adhesion strength of the coating shall not fall below 10 lb per inch of width after seawater immersion as specified in 4.6.2.1.6.

3.5.3.8 Temperature resistance. The boom material shall remain serviceable as an oil containment boom over a temperature range of -40° Fahrenheit (F) to +150°F, as evidence by no cracks in a 180° angle bend test in accordance with 4.6.2.1.7, at a temperature of -40°F. Further, the outer covering of the material shall not become tacky when heated to +150°F. The boom material shall not sag at a temperature of +150°F when oriented vertically and supported in the same manner as in the assembled boom.

3.5.4 Flotation. The following paragraphs pertain to the durability requirements of boom flotation and apply only to the flotation protective coating.

3.5.4.1 Puncture resistance. When tested in accordance with 4.6.2.1.3, the flotation coating at its thinnest point shall have a puncture strength of no less than 150 lb.

3.5.4.2 Abrasion resistance. After being subjected to the abrasion conditions specified in 4.6.2.1.4, the flotation coating shall exhibit a puncture strength of no less than 125 lb when tested in accordance with 4.6.2.1.3.

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3.5.4.3 Weathering resistance. The flotation coating shall be subjected to the weathering conditions of 4.6.2.1.5. The puncture strength of a flotation coating test sample shall be no less than 125 lb when tested in accordance with 4.6.2.1.3. The puncture strength of the flotation coating test sample, when weathering and abrasion are combined, shall be no less than 100 lb when tested in accordance with 4.6.2.1.3.

3.5.4.4 Water absorption. The flotation shall be hermetically sealed to prevent water takeup when the boom is continuously immersed in salt, brackish, or fresh water.

3.5.4.5 Temperature resistance. The flotation coating shall remain serviceable for the purposes of an oil containment boom as evidenced by no cracks after the 180° bend test as specified in 4.6.2.1.7 at a temperature of -40°F and no tackiness of the coating surface at a temperature of +150°F.

3.5.4.6 Boom attachment method. All mechanical attachment fittings between flotation and boom material shall be designed to minimize stress concentrations which may result in excessive abrasion or tearing.

3.5.5 Handholds. Each boom section shall have no less than 10 handholds between the end connectors. Each handhold shall be capable of sustaining a 300 lb load applied as specified in 4.6.2.1.8 without permanent set, peeling, tearing, or elongation which cause any loss in serviceability.

3.5.6 Ballast. If required to maintain a vertical attitude under the conditions of 3.6.3, ballast weight shall be located at the bottom of the boom draft. The ballast material shall be integral with the boom material in such a way as to preclude relative motion of the ballast and boom material (to insure that the ballast member does not abrade the boom material). The ballast member shall not be attached to the Navy standard boom section end connectors, so that all tensile loads on the boom must be borne by the boom material itself as specified in 3.5.8.

3.5.7 Load lines. For the purpose of minimizing boom abrasion, puncture and tears, and to minimize vessel down time should the boom accidently be drawn into a ship's propellers, no chains or cables shall be used to achieve the strength requirements of 3.5.8. The total tensile requirements of 3.5.8 shall be satisfied by the barrier material alone.

3.5.8 Structural strength. The boom shall be designed and constructed to withstand the induced forces resulting from the straight line tow maneuvers and a catenary configuration against the current as specified in 3.6.1 and 3.6.3, but in any case, shall have an assembled ultimate tensile strength of no less than 350 lb per inch of boom height.

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Any damage to the end connectors, floats, or their boom attachment points or to the material itself during the first article test will be cause for rejection. The strength of the boom assembly and the lack of tears due to relative movement between floats and other attachments and the boom material shall be verified by subjecting a 15 foot test length, complete with floats and standard Navy end connectors, to the tensile loading test as specified in 4.6.2.1.10.

3.5.9 Hardware. Hardware shall be as specified on Dwg No. 73-11-1F through 73-11-4F.

3.5.9.1 Boom connectors. Boom connectors, as specified in 3.5.9 shall be supplied with each boom section. Each boom section shall have a female connector on one end and a male connector on the other end (see NAVFAC Dwg 73-11-1F).

3.5.9.1.1 Anchor points. Anchor points in the form of eye bolts and eye nuts located near the top and bottom of the standard Navy end connectors shall be 1/2 inch in size vice the 3/8-inch size shown on the boom connector drawing. The clearance holes (top and bottom) shall be sized to the 1/2-inch eyebolt (see NAVFAC Dwg 73-11-1F). The minimum tensile strength of the anchor points shall be 10,000 pounds per square inch (psi) and the minimum shear strength shall be 5,000 psi.

3.5.9.2 Boom connector adapter. Boom connector adapters (see NAVFAC Dgw 73-11-2F), as specified in 3.5.9, shall be supplied in the amount specified in the contract.

3.5.9.3 Boom bulkhead assembly. Boom bulkhead assemblies (see NAVFAC Dwg 73-11-3F) as specified in 3.5.9, shall be supplied in the amount specified in the contract.

3.5.9.4 Screw threads. All screw threads shall be in accordance with NBS H28.

3.5.10 Stability/tension line. To satisfy the stability requirements of 3.6.3, a stability/tension line may be provided. It may be synthetic line or webbing, metallic line is not allowed, and shall run the entire length of each boom section terminating at the section end connector.

3.5.10.1 Stability/tension line fasteners. When a stability line is used, the method of fastening the line to the boom shall be determined by contractor design, but the mechanical and structural integrity of the

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system shall be capable of withstanding the forces imposed on the stability line and shall not impair boom deployment. This line, when used, shall be affixed to the boom in a manner so as not to dangle, loop, or sag creating a potential to entrap debris or to become an entanglement hazard.

3.5.11 Repair kit. A field repair kit shall be provided in the quantities specified in the contract (see 6.2.1). The contents of the repair kit, subject to the approval of the Government shall include, as a minimum, the following items:

(a) When external flotation is used, sufficient repair flotation elements to restore one 50 foot section of boom, complete with all necessary mounting tools and materials necessary to mount the elements onto the boom. When integral flotation is supplied, a continuous 50 foot length of boom and its flotation.
(b) A 50 foot length of replacement boom material, with required quantity of flotation elements, to repair cut or damaged sections of boom.
(c) Ten boom handle assemblies, complete with tools and materials necessary for their attachment to a boom section whose handholds have been lost in service.
(d) Repair instruction pamphlet containing adequate diagrams and instructions for:

(1) Remaking end connections with the standard Navy end connectors on completely severed boom sections.
(2) Replacing lost or damaged ballast members.
(3) Repairing tears and/or punctures in floats, boom material, and boom material adjacent to end connectors.
(4) Replacing floats, handholds, and anchor points.
(5) Instructions on cleaning and preparing surfaces prior to repair.

(e) Replacement ballast for 50 feet of boom.
(f) Four sets of replacement anchor points.
(g) One set of Navy Standard end connectors (male and female).

3.6 Performance.

3.6.1 Operational. The boom shall be capable of containing, enclosing, and aiding in cleanup operations of limited oil spills in dockside areas surrounding vessels, barges and waste oil rings, and other harbor craft. The boom shall be easily coupled and decoupled, when under strain due to prevailing currents, using the special connector adaptors described in 3.5.9.2. The usual deployment of the boom is from a rafted configuration to allow for vessel movement to a configuration, depending upon local geography, to enclose the high risk spill area around docked vessels. Accordingly, the boom must be capable of sustaining the towing maneuver of 4.6.2.2.2 without damage to the flotation, boom material, or loss of floats.

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3.6.2 Flexibility. The boom must have sufficient flexibility and articulation along its length to conform to the water surface and not bridge from crest to crest waves having a height of 8 feet and wave length of not more than 50 feet. It must also have sufficient flexibility to allow rafting in a storage position as specified in 4.6.2.2.2.

3.6.3 Stability. The boom shall have a static righting moment of no less than 6.0 foot lb per foot when tested as specified in 4.6.2.1.15 and a skirt deflection from the vertical of no more than 30° when subjected to a current of 3 knots as specified in 4.6.2.2.3. Furthermore, a minimum of eight vertical stiffeners (in addition to end connectors) running the full height of the boom (bottom of barrier skirt to top of barrier fence) shall be used for each 50 foot section. In lieu of stiffeners, the boom design shall permit no more than 2 inch loss in freeboard when subjected to the test of 4.6.2.2.1.

3.6.4 Bumpering forces. The boom shall be so constructed as to withstand the drop test of 4.6.2.1.14 without any permanent structural damage to flotation elements or any other part of the boom test sample which would affect long term serviceability.

3.6.5 Field repairability. The boom shall be field repairable. Repairs which must be conducted (see 4.6.2.2.4) are: replacement of flotation elements and assemblies; repairs to tears in the boom freeboard or draft and the replacement of a standard Navy end connector on a freshly trimmed damaged boom end. Permanent repairs shall not require the use of tools or procedures beyond those found in the repair kit as specified in 3.5.11.

3.6.6 Operational replacement of sections. When tested as specified in 4.6.2.2.5, a 50 foot section of boom shall be replaced while at least three sections of boom are deployed.

3.6.7 Cleaning. Due to its continual immersion in salt, brackish, or fresh water, marine growth on the boom will unavoidably occur. Cleaning this growth from the boom while in the water will typically consist of mechanical scraping means. These mechanical means shall not degrade the abrasion, puncture, or tensile strength requirements of 3.5.3. In addition, once out of the water, the boom shall be cleanable of oil residues, as verified by the procedure of 4.6.2.1.12, using nontoxic industrial cleaning solutions. Any permanent softening or dissolving of the boom material or flotation coating following this operation shall be cause for rejection.

3.7 Construction. The equipment shall be designed and constructed to facilitate field maintenance. All adjustments and replaceable accessories shall be readily accessible. Conditions which can be hazardous to personnel or deleterious to equipment shall not be permitted.

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3.7.1 Hardware. Construction or manufacturing processes shall be as specified herein or on the applicable drawings.

3.8 Dissimilar metals. Intimate contact which can be expected to cause galvanic corrosion shall be avoided. When such contact cannot be avoided, an interposing insulating material shall be provided to minimize the corrosive effect.

3.9 Color. The color of the boom and its major components shall conform to FED STD No. 595, approximating color No. 12197 international orange.

3.10 Workmanship. The boom, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp edges; accuracy of dimensions, and marking of parts and assemblies; completeness of coating, sewing, soldering-welding, brazing and riveting; alinement of parts and tightness of assembly screws and bolts; or any other defects which could impair serviceability.

3.11 Technical publications. Technical publications shall be as specified in the contract and shall include instructions for connecting and deploying the boom and guidelines for installation and utilization of the boom system, plus such other instructions that are required to inspect, maintain, clean, and repair the boom (see 3.5.11 and 6.3). The number of copies to be furnished shall be as specified (see 6.2.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the contractor is responsible for insuring that materials and components used were manufactured, examined, and tested to the extent specified, in accordance with the requirements of referenced specifications and standards.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be performed on one complete barrier system when a first article sample is required (see 3.2 and 6.2.1). This inspection shall include the first article test plan of 4.6.1, the examination of 4.5 and the tests of 4.6.2. Failure of the first article to pass the examination or any of the tests shall be cause for rejection. The first article may be a standard production item from the contractor's current inventory provided the unit meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining units to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on the barrier as specified in 4.6.3.

4.3 Inspection lot. All units offered to the Government at one time shall be considered a lot for purposes of inspection.

4.4 Sampling.

4.4.1 Sampling for examination.

4.4.1.1 Boom with end connectors. Every tenth section of boom with the end connectors attached shall be examined in accordance with 4.5.

4.4.1.2 Hardware. One sample of each unit of hardware shall be randomly selected from each lot and examined in accordance with 4.5.

4.4.2 Sampling for test. Samples of the boom material to be used for the tests of 4.6.2 shall be selected as in 4.4.2.1 and 4.4.2.2. Samples of the flotation coating to be subjected to the tests of 4.6.2 shall be selected as in 4.4.2.3.

4.4.2.1 Boom material. Two random samples of the boom freeboard and skirt material shall be selected from each 500 feet of boom length or fraction thereof. The sample shall be no less than 1-1/2 feet long and the full depth of the boom including freeboard and skirt. The sample shall be identical in all manufacturing respects, e.g., thickness and method of application of coating or impregnated covering, and chemical additives with the material used in production boom lengths.

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4.4.2.2 Boom material seams/splices. Where material seams or splices are used between 50 foot section ends, samples of the seam/splices must be obtained for the tests of 4.6.2.1.9 in the following way: Not less than three samples of each type of seam/splice shall be furnished for tests. Samples shall be the same as used in the production of boom lengths, and shall be fabricated and bonded identically and simultaneously with the boom seams/splices. The sample shall have at least 4 inches of parent material on each side of the seam/splice and shall have a length equal to the boom depth. The seam/splice shall be centered in all samples.

4.4.2.3 Flotation coating. Samples of the flotation coating to be subjected to the applicable tests of 4.6.2.1 shall be obtained by cutting a 6 inch by 6 inch flat square sample of the coating from the thinnest section of a flotation element identical to the production items to be supplied under this contract. The flotation foam shall be carefully removed from the underside of the sample before testing as specified in 4.6.2.1.

4.5 Examination. The first article and each sample selected in accordance with 4.4.1 shall be examined for compliance with the requirements of Section 3 of this specification. The examination shall be conducted as specified in Table I. Any sample having one major defect or two minor defects shall constitute cause for rejection.

TABLE I. List of defects.

Categories	Defects
Major:	
101	Material not as specified.
102	Units of same size not interchangeable.
103	Abraded or torn material.
104	Material coating separation.
105	Defective material seams.
106	Missing or damaged components.
107	Design and construction not as specified; dimensions not as specified.
108	Workmanship is inferior and not as specified; extrusions not free from burrs and sharp edges; bolt holes not accurately drilled to coincide with bolts or other fittings.
109	Surfaces not cleaned or treated as specified.
110	Repair kit not as specified. Components missing or damaged.
111	Loose or unfastened flotation or ballast.
Minor:	
201	Marking omitted or incorrect.
202	Color of boom not as specified.
203	Loose handholds.

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to insure compliance with 3.5.3.1. When other than coated or impregnated woven fabric is used for the boom material, a similar test of ultimate tensile strength shall be performed for samples in two orthogonal directions (regardless of the elongation of the sample).

4.6.2.1.2 Tearing strength (boom material). When coated fabric is used as the boom material the tearing strength shall be determined for conformance with 3.5.3.2 in accordance with Method 5134 of FED TEST METHOD STD. No. 191. The tear strength in both the warp and fill directions shall be measured. When other than coated fabrics are used for boom material, a similar test of tear strength in two orthogonal directions shall be performed to insure compliance with 3.5.3.2.

4.6.2.1.3 Puncture strength (boom material and flotation coating). Samples of boom material as specified in 4.4.2.1 and flotation coating as specified in 4.4.2.3 shall be fastened in a specimen holder in accordance with Figure 3. A piercing instrument with its end conforming to Figure 3 shall be forced against the material at approximately the center of the area enclosed by the specimen holder. The rate of travel of the piercing instrument shall not exceed 20 inches per minute. The force required to puncture the material shall be not less than specified. The results of five individual tests shall be averaged and used to insure compliance with 3.5.3.3 for boom material and 3.5.4.1 for flotation coating.

4.6.2.1.4 Abrasion resistance (boom material and flotation coating). Samples of boom material as specified in 4.4.2.1 and flotation coating as specified in 4.4.2.3 shall be subjected to the following abrasion condition in accordance with Method 5304.1 of FED TEST METHOD STD. No. 191. Five flotation coating samples shall be abraded on their exterior surface for 2,000 cycles with 80 grit garnet paper under a 2 lb weight. Twenty-five boom material samples shall be abraded in a direction parallel to the boom length on both sides and 25 boom material samples shall be abraded in a direction perpendicular to the boom length on both sides for 1,000 cycles each with 80 grit garnet paper under a 2 lb weight. NOTE: The garnet paper used for testing under this paragraph shall be replaced with new, unused 80 grit garnet paper every 200 cycles. Tests, number of samples per test, and test result requirements are listed in Table II.

4.6.2.1.5 Weathering resistance (boom material and flotation coating). Samples of boom material as specified in 4.4.2.1 and flotation coating as specified in 4.4.2.3 shall be subjected to the following accelerated weathering conditions in accordance with Method 5804.1 of FED TEST METHOD STD. No. 191. Seventy-five samples of the boom material and 10 samples of the flotation coating shall be subjected to the weathering of an alternating carbon arc and water spray condition extending for 150 hours. The tests listed under "weathering" in Table II shall then be performed on the required number of samples of boom material and flotation coating. The remaining samples of boom material and flotation coating shall be subjected to the abrasion conditions of 4.6.2.1.4 and tested as listed under "combined weathering and abrasion" in Table II.

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1.5.1 and ASTM D 996 shall apply.

Table II. Tests and requirements.

	Test	Ref. Paragraph	Special instructions	No. of samples each Load direction			Minimum Requirements	
				Warp	Fill	Puncture	Req'ts	Ref. Para
BOOM MAT'L	I ABRASION RESISTANCE	4.6.2.1.4						
	(A) TENSILE	4.6.2.1.1	ABRADED PARALLEL TO BOOM	5	5	-	400 lb	3.5.3.4
	(B) TEAR	4.6.2.1.2	ABRADED PARALLEL TO BOOM	5	5	-	200 lb	3.5.3.4
	(C) PUNCTURE	4.6.2.1.3	ABRADED PARALLEL TO BOOM	-	-	5	125 lb	3.5.3.4
	(D) TENSILE	4.6.2.1.1	ABRADED PERPENDICULAR TO BOOM	5	5	-	400 lb	3.5.3.4
	(E) TEAR	4.6.2.1.2	ABRADED PERPENDICULAR TO BOOM	5	5	-	200 lb	3.5.3.4
	(F) PUNCTURE	4.6.2.1.3	ABRADED PERPENDICULAR TO BOOM	-	-	5	125 lb	3.5.3.4
	II WEATHERING RESISTANCE	4.6.2.1.5						
	(A) TENSILE	4.6.2.1.1		5	5	-	400 lb	3.5.3.5
	(B) TEAR	4.6.2.1.2		5	5	-	200 lb	3.5.3.5
	(C) PUNCTURE	4.6.2.1.3		-	-	5	125 lb	3.5.3.5
	III COMBINED WEATHERING & ABRASION RESISTANCE	4.6.2.1.5						
	(A) TENSILE	4.6.2.1.1	ABRADED PARALLEL TO BOOM	5	5	-	375 lb	3.5.3.5
	(B) TEAR	4.6.2.1.2	ABRADED PARALLEL TO BOOM	5	5	-	190 lb	3.5.3.5
	(C) PUNCTURE	4.6.2.1.3	ABRADED PARALLEL TO BOOM	-	-	5	100 lb	3.5.3.5
	(D) TENSILE	4.6.2.1.1	ABRADED PERPENDICULAR TO BOOM	5	5	-	375 lb	3.5.3.5
	(E) TEAR	4.6.2.1.2	ABRADED PERPENDICULAR TO BOOM	5	5	-	190 lb	3.5.3.5
	(F) PUNCTURE	4.6.2.1.3	ABRADED PERPENDICULAR TO BOOM	-	-	5	100 lb	3.5.3.5
	IV WATER ABSORPTION	4.6.2.1.6						
	(A) TENSILE	4.6.2.1.1		5	5	-	500 lb	3.5.3.7
	(B) TEAR	4.6.2.1.2		5	5	-	250 lb	3.5.3.7
	(C) PUNCTURE	4.6.2.1.3		-	-	5	150 lb	3.5.3.7
	V ADHESION OF COATING	4.6.2.1.6	WHEN REQUIRED, 5 SAMPLES SHALL BE TESTED, AVERAGED, AND REPORTED AS SPECIFIED IN 3.5.3.7	-	-	-	10 lb/in	3.5.3.7
FLOTATION MAT'L	I ABRASION	4.6.2.1.4	PUNCTURE TEST ONLY	-	-	5	125 lb	3.5.4.2
	II WEATHERING	4.6.2.1.5	PUNCTURE TEST ONLY	-	-	5	125 lb	3.5.4.3
	III COMBINED WEATHERING & ABRASION	4.6.2.1.5	PUNCTURE TEST ONLY	-	-	5	100 lb	3.5.4.3

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4.6.2.1.6 Water absorption (boom material). Twenty-five samples of boom material as specified in 4.4.2.1 shall be submerged in 70°F synthetic seawater conforming to ASTM D1141 for a continuous period of 20 days. The 25 samples shall then be subjected to the tensile test of 4.6.2.1.1, the tear test of 4.6.2.1.2, and puncture test of 4.6.2.1.3. The test results from each of the five tensile, five tear, and five puncture tests shall be averaged and reported as specified in 3.5.3.7. In addition, when the boom material used is a coated fabric of the type upon which an adhesion of coating test can be conducted without destroying the fabric, a test of the adhesive strength of the coating as specified in Method 5970 of FED TEST METHOD STD. No. 191 shall be measured and reported as required in 3.5.3.7.

4.6.2.1.7 Temperature resistance (boom material and flotation coating). A 1 by 4 inch specimen of the coated cloth with the long dimension warpwise and a 1 by 4 inch specimen of the coated cloth with the long dimension fillingwise shall be exposed for 4 hours at a temperature of -40° ±5°F with the temperature recorded at the lowest point in the chamber. The sample shall then be bent 180° over a 1-inch steel pipe that has been exposed in the test chamber with the test specimen. When heated to a temperature of +150°F, the surfaces of both boom material and flotation coating samples shall not become tacky or sticky.

4.6.2.1.8 Handhold strength test. One 50 foot long boom section, identical to those to be delivered, shall be subjected to a test to insure compliance with 3.5.5. Three of the handholds on the boom section shall be subjected to a 500 lb load for a duration of 15 minutes in each of the following directions: along the boom length, horizontally at right angles to the boom length and vertically at right angles to the boom length. The load shall be applied with a 3/4-inch rope loop through the handhold. A visual inspection following this test shall be made to determine compliance with 3.5.5.

4.6.2.1.9 Seam/splice breaking strength. Seam/splice samples shall be selected as specified in 4.4.2.2 with the seam parallel to the short dimension and centered. The edges of the specimen shall not be covered or coated in any additional manner. The breaking strength of the seam sample shall be determined as specified in Method 5100.1 of FED TEST METHOD STD. No. 191 and reported in lb to determine compliance with 3.5.3.6.

4.6.2.1.10 Assembled boom tensile strength. A test length of assembled boom, complete with flotation, ballast (if used) and standard Navy end connectors and having a length no less than 15 feet, shall be subjected to a uniform tensile load over the height of the boom of 350 lb per inch of boom height and held constant for 15 minutes. Stability/tension lines shall be disconnected from end connectors during this test. At the end of this period, a visual inspection shall be made of the boom assembly including boom material, Navy end connector joints, and float attachment points to insure compliance with 3.5.8.

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4.6.2.1.11 Boom weight. At least three 50 foot sections of boom shall be selected at random and weighed to determine conformance to 3.5.2.5.

4.6.2.1.12 Cleaning. Place a completely assembled boom section 50 feet in length in an area that is suitable for performing a cleaning operation. Place and spread completely on approximately 10 feet of one side of the test section Navy Special Fuel Oil conforming to MIL-F-859 or diesel fuel oil, grade 2-D, (ASTM-D975) in a sufficient quantity to simulate the condition of the boom after operational use during an oil spill. Utilizing contractor's recommended nontoxic cleaning materials, thoroughly clean the oil from the test section to demonstrate conformance to 3.6.7. Specific notation shall be made of those areas of the boom requiring special attention during cleaning to insure complete oil removal.

4.6.2.1.13 Resistance to hydrocarbons. A 2-inch by 3-inch sample of the fabric and the flotation coating shall be immersed in the test fluid of 4.6.2.1.12 for 70 hours (±1 hour) at 23°Celsius (C) (±2°C). The tested samples shall be deemed unacceptable if they crack, soften and become sticky, or otherwise show signs of degradation which would affect long term serviceability.

4.6.2.1.14 Bumpering forces. Construct a test section of the boom complete with a single flotation element but having a length no less than 2 feet. Place the test section on a solid floor with the boom freeboard and skirt held parallel to the floor, as specified in Figure 4. A weight having a 2 foot by 2 foot steel plate as its bottom surface shall be suspended above the boom test section as specified in Figure 4. The weight and drop height measured from the highest point of the boom flotation shall be such as to equal a potential energy of no less than 2,280 foot lb. The weight shall be released and allowed to impact the boom flotation. A visual inspection of the test section, particularly the flotation element, shall be conducted to determine compliance with 3.6.4.

4.6.2.1.15 Static righting moment. Construct a test section of boom, complete with one flotation element and ballast (if used). The ballast shall have a value per foot equal to that of the assembled boom. The test section shall have a length no less than 2 feet. Place this test section in a fresh water tank of sufficient dimension to allow the section to remain clear of any surface during the course of the following procedure. Attach hardware, as necessary, to incline the section to an angle 30° from the vertical about the waterline axis and to measure the required moment as shown in Figure 5. Incline the test section about its waterline to an angle with the vertical of 30° and measure the resulting moment, in foot lb, necessary to do this. Determine the static righting moment per foot of assembled boom by the equation:

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$$\overline{M} = \frac{M}{L_1} \quad f \qquad (1)$$

Where: $\overline{M}$ = static righting moment per foot assembled boom, foot lb per foot.
M = moment necessary to incline test section 30°, foot lb.
L_1 = length of test section, foot.
f = multiplying factor = $\frac{N\ L_2}{100\ \text{foot}}$

Where: N = number of test section floats in 100 foot of assembled boom length.
L_2 = length of test section float, feet.

For example, if there are 22 test section floats, each 2.5 feet long in a 100 foot length of assembled boom, then:

$$F = \frac{(22)\ (2.5)}{100} = 0.55$$

Similarly, if continuous flotation is used in the assembled boom and in a test section length of say 2 feet, then:

$$f = \frac{(50)\ (2)}{100} = 1.0$$

The static righting moment per foot of assembled boom length shall be determined as in equation (1) above and reported to determine compliance with 3.6.3.

4.6.2.1.16 Hardware tensile load test. Each unit of hardware specified in 4.5 shall be subjected to the following tests as applicable. In addition, the flotation shell of the boom tow assembly shall be tested as specified in 4.6.2.1.17.

(a) Boom connector. Assemble one boom connector pair (male and female) and subject the assembly to a uniform test load of 22,500 lb. Deformation of the male or female connectors shall be cause for rejection. Deformation is defined as a change in linear measurement between two points, punchmarked on the semicircular portion of the male and female connectors prior to testing, on a line drawn perpendicular to the direction the tensile load is imposed.

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(b) Boom connector adapter. Same as for the male portion of the boom connector.

(c) Boom bulkhead and boom tow assemblies. The boom bulkhead and boom tow assemblies shall be subjected to the tensile load tests specified in (a). In addition to the deformation rejection criteria specified any visual distortion or structural weld failure shall be cause for rejection.

4.6.2.1.17 Water-tight integrity test. The flotation shell of the boom tow assembly shall be tested for water-tight integrity by chilling the shell to a temperature of 50 ±10°F and immersing the shell in water with a temperature of 130 +10°F. The shell shall be kept submerged for not less than 3 minutes. Bubbling from the shell, indicating leakage, shall be cause for rejection. This test shall be conducted after the shell has been filled with polyethylene foam and sealed.

4.6.2.1.18 Locking rod test. An initial tension load shall be applied to the boom to lock the rod into the end connector. A 15 lb axial load shall then be applied to the end of the locking rod for 60 seconds. Relative slippage between the end connector and locking rod shall be not greater than 10 percent of the length of the connector when subjected to this test.

4.6.2.2 Field first article tests. The boom first article and its components shall be subjected to the field tests specified in 4.6.2.2.1 through 4.6.2.2.5. Failure to meet the minimum acceptable test results listed in the referenced paragraphs in Section 3, of this specification shall be cause for the Government to refuse acceptance of all booms until corrective action has been taken and all the minimum acceptable test results are met. Tests shall be conducted by the contractor and in the presence of Government representatives. All of the following tests shall be conducted at a harbor test site in accordance with a test plan approved by the Government as specified in 4.6.1.

4.6.2.2.1 Freeboard and draft measurement. A 50 foot long boom test section shall be placed in the water and secured in such a way that it assumes a free floating attitude (no tensile forces acting on boom). Direct measurement of the boom freeboard shall be made at no fewer than five equally spaced locations along the boom. The boom draft at each of these points shall be determined by subtracting the freeboard measurements from the overall height of the boom at each of these points. The average values of the draft and freeboard measurements obtained in this manner shall be used to determine compliance with 3.5.2.1 and 3.5.2.2. Should vertical stiffeners not be incorporated into the design, the 50 foot boom test

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section shall be such that when subjected to a 25 knot wind perpendicular to its length (as measured by an air speed indicating device) loss in freeboard is not in excess of that stipulated in 3.6.3. Noncompliance with this requirement shall constitute failure of the test.

4.6.2.2.2 Single end towing maneuvers. Ten sections of the assembled boom, forming a 500 foot test length, shall be placed in the water in a rafted position as specified in Figure 6A and tied off to an adjacent pier structure to simulate temporary storage of permanent boom during vessel movement operations. With one end of the 500 foot test length secured to the pier structure, a suitable two-leg towing bridle shall be attached to the anchor points on the standard Navy connector at the opposite end of the test length. A harbor craft of sufficient power shall be secured to the towing bridle and used to tow the rafted boom into a straight line, simulating deployment of the boom in an oil containment position. The tow shall proceed at a speed of no less than 1 knot. Visual observation during an inspection after this procedure shall be used to determine compliance with 3.6.1. In a second towing maneuver test the 500 foot test length shall be towed into a straight line and then in a reverse direction at 180° to the original towing direction as specified in Figure 6B. A visual inspection shall be used to determine compliance with 3.6.1.

4.6.2.2.3 Roll stability in current. The following test shall be performed to determine compliance with the stability requirements of 3.6.3. Ten sections of the assembled boom, forming a 500 foot test length of fully assembled boom, shall be subjected to a relative water velocity in the catenary configuration shown in Figure 7. Either towing with towboats or deployment of the test section in a region of the appropriate current may be utilized to obtain the required water velocity relative to the test boom length. The distance across the opening of the catenary shall measure 450 feet. Instrumentation to measure boom skirt inclination shall be attached to the boom as shown at the bottom of the catenary on its centerline and at 50 feet on either side of the catenary centerline. The inclination of the boom skirt from the vertical shall be read directly from the instrumentation attached to the boom skirt at these selected points. Measurements shall be obtained at a water velocity relative to the boom of 3 knots. The three angle measurements of the boom shall be averaged to determine compliance with 3.6.3.

4.6.2.2.4 Field repairability. To demonstrate compliance with 3.6.5, the following operations shall be performed, with the repair kit of 3.5.11, on a representative 10 foot length of assembled boom. For those booms having flotation that is not integral with the boom material, one complete float assembly on both sides of the boom shall be removed and replaced. Regardless of the manner of flotation attachment, a cut shall be made in the boom freeboard and skirt with a knife or other suitable tool. These tears shall then be patched sufficiently to yield a leak

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tight condition. The standard Navy connector shall be completely severed from one boom end and replaced after trimming and preparing the severed boom material end as necessary.

4.6.2.2.5 Operational replacement of section test. Three sections of the assembled boom, forming a 150 foot test length, shall be placed in the water and secured in such a way that it assumes a free floating attitude. The middle section of the boom shall be removed and replaced by men working from a small boat.

4.6.2.3 Repair kit. Unless otherwise specified (see 6.2.1), the contractor shall demonstrate the ability of the repair kit to make the repairs listed in 3.5.11 (a through g).

4.6.3 Quality conformance tests. The contractor shall submit for Government approval, a quality assurance plan for production of permanent booms procured under the contract. The plan shall insure that all booms and hardware procured under the contract meet all of the performance requirements specified herein. The plan shall, as a minimum, outline the sampling method, frequency and tests of the boom material, flotation elements and attachment hardware to insure quality control during production. The tests shall include, as a minimum, 4.6.2.1.1 through 4.6.2.1.3, 4.6.2.1.8, 4.6.2.1.10, 4.6.2.1.15 and 4.6.2.1.16. Unless otherwise specified (see 6.2.1), verification of contractor filed Certificate of Compliance shall be accepted in lieu of quality conformance testing.

4.7 Packaging inspection. The preservation-packaging, packing, and marking of the barrier system shall be inspected to verify conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation-packaging. The preservation-packaging shall be level A or C, as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Disassembly. Disassembly shall be the minimum necessary to safeguard components known to be subject to damage or loss and to accomplish reduction in cube. Bolts, nuts, pins, and washers removed shall be reinstalled in one of the mating parts and secured to prevent their loss.

5.1.1.2 Booms. The booms shall be folded, rolled or coiled, as applicable, to a minimum safe diameter, and secured if required, with cotton cord or by other suitable means.

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5.1.1.3 Hardware. Hardware, which is not attached to the boom, shall be packaged in accordance with MIL-P-116, Method III.

5.1.1.4 Repair kit. Components of the repair kit shall be packaged in accordance with MIL-P-116, Method III.

5.1.1.5 Instruction pamphlets. The instruction pamphlet shall be packaged Method IC-3 in accordance with MIL-P-116.

5.1.1.6 Consolidation packaging. Small components of the repair kit and hardware shall be consolidated packaged in close-fitting boxes conforming to PPP-B-636, class weather-resistant. The contents shall be blocked, braced, and cushioned to prevent movement inside the container or damage to the contents. The boxes shall be closed in accordance with the appendix to the box specification, Method IV.

5.1.2 Level C. The complete barrier system shall be preserved and packaged in a manner that will insure protection against deterioration and damage during shipment.

5.2 Packing. Packing shall be level A or C, as specified (see 6.2.1).

5.2.1 Level A. The barrier system shall be packed in boxes conforming to PPP-B-601, overseas type. The contents shall be blocked and braced to prevent movement.

5.2.2 Level C. The barrier system shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules, or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required in the contract, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The permanent containment boom specified herein is intended for use in containing, enclosing, and aiding in cleanup operations of oil spills in dockside areas surrounding vessels, barges, waste oil rings, and other harbor craft.

6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements:

(a) Title, number, and date of this specification.
(b) Type of hardware required (see 1.2).

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(c) When a first article sample is required for inspection and approval (see 3.2, 3.2.1, 4.2.1, and 6.4).
(d) When a first article test plan required (see 3.2.2, 4.6.1, and 6.4).
(e) Quantity of technical publications (see 3.11).
(f) Quantity of repair kits required (see 3.5.11).
(g) When first article testing of repair kit is not required (see 4.6.2.3).
(h) When Certificate of Compliance shall not be used in lieu of quality conformance tests (see 4.6.3).
(i) Level of preservation-packaging and level of packing required (see 5.1 and 5.2).

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulation (ASPR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data specified below shall be delivered in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirements	Applicable DD 1664
3.2.1 and 4.6.1	Test Plan, First Article Qualification	DI-T-5315A
3.5.11(d) and 3.11	Publications, Commercial	DI-M-24010A

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the ASPR. The first article should be a first production item consisting of one complete boom with hardware, container, and repair kit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

6.5 Definitions. Referring to Figure 2, the terminology used in this specification is as follows:

(a) Boom - synonym for floating barrier with end connectors and flotation; a section or series of connected sections.
(b) Boom freeboard - that portion of the boom extending above the still water line designed to prevent loss of contained oil due to splashover (see 3.5.2.1).

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(c) Boom draft - that portion of the boom extending below the still water line designed to prevent loss of contained oil due to oil drainage or oil particle entrainment in response to water currents directed perpendicular to the boom (see 3.5.2.2).
(d) Boom flotation - solid foam flotation attached to the boom necessary to maintain adequate freeboard in response to marine growth, waves, wind, or water current relative to the boom (see 3.5.2.3).
(e) Boom section - a continuous length of boom 50 feet in length having a female Navy connector on one end and a male Navy connector on the other end (see 3.5.2.4).
(f) Boom end connectors - Navy standard mechanical connectors to allow decoupling of boom sections after the boom is deployed in the water (see 3.5.9.1).
(g) Bulkhead attachment assembly - Navy standard design to provide a leak tight method for connecting the boom to a pier or dock structure under tidal conditions (see 3.5.9.3).
(h) Handholds - fixtures on the boom freeboard to assist handling and positioning the boom in the water (see 3.5.5).
(i) Anchor points - eye bolts located near the top and bottom of the standard Navy connector at each boom section end for the purpose of mooring the boom in various configurations (see 3.5.9.1.1).
(j) Boom connector adapter - special adapters (see 3.5.9.2), which allow decoupling of boom sections under tensile load.
(k) Stability/tension line - a supplementary tension member to aid in meeting stability requirements.

Preparing activity:

Navy - YD

Project No. 1945-N020

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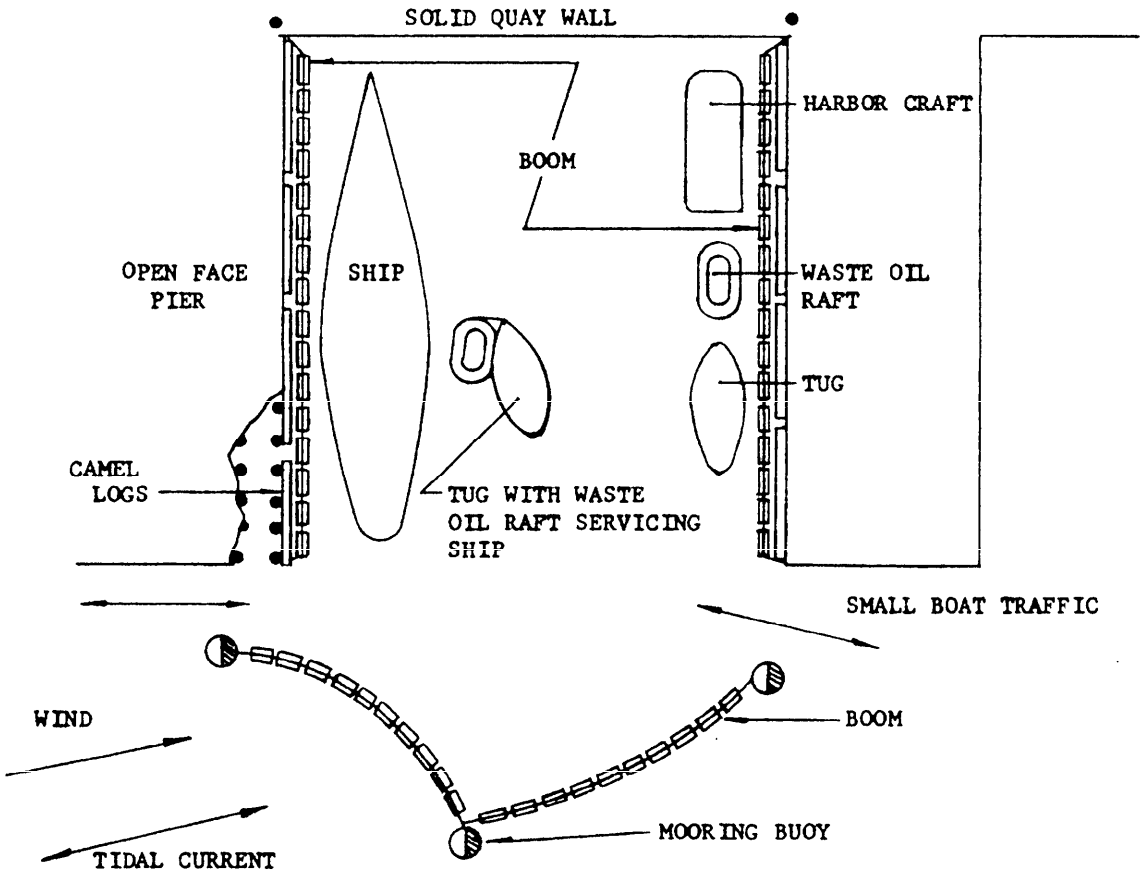


FIGURE 1 - USE OF BOOM

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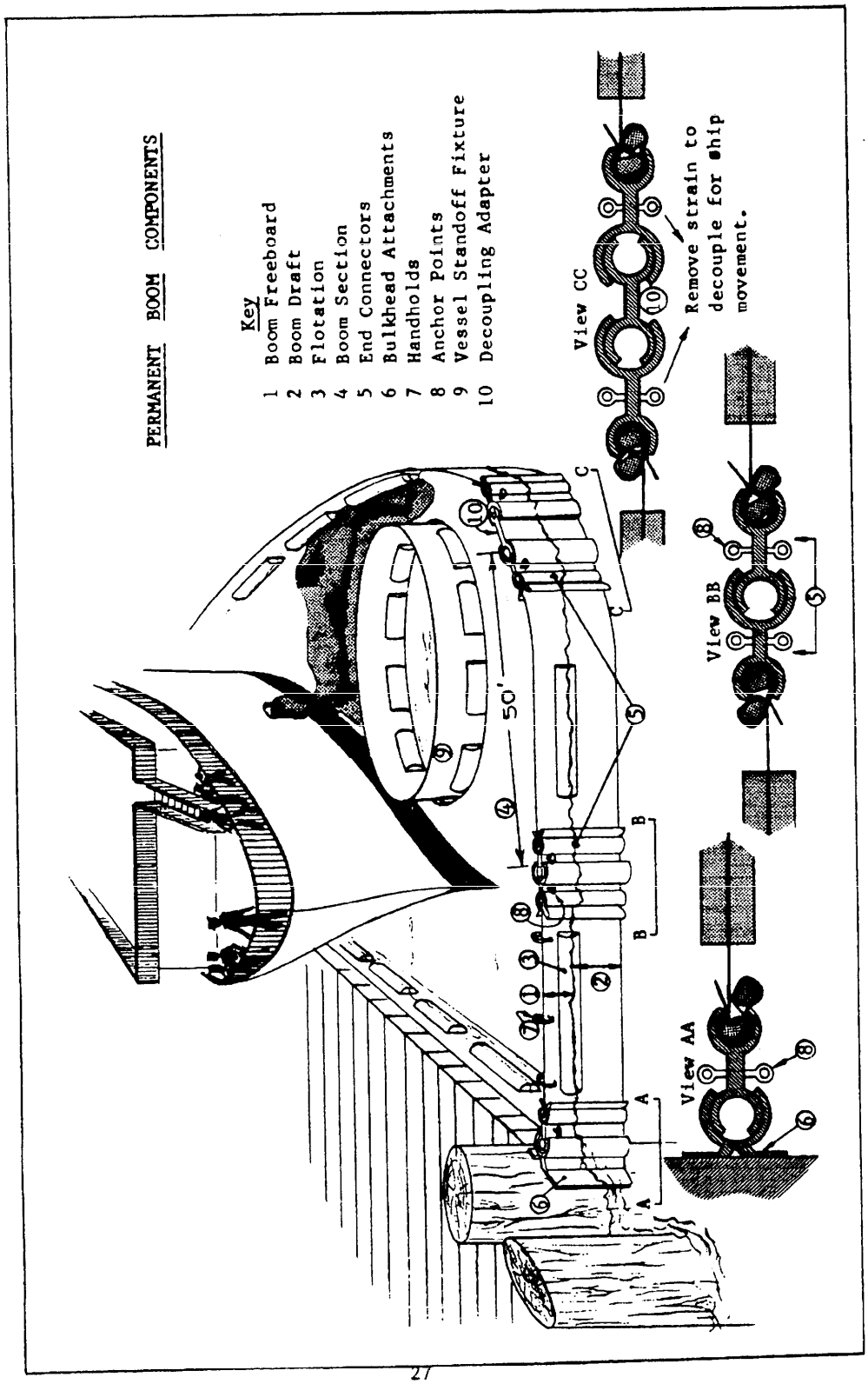


FIGURE 2

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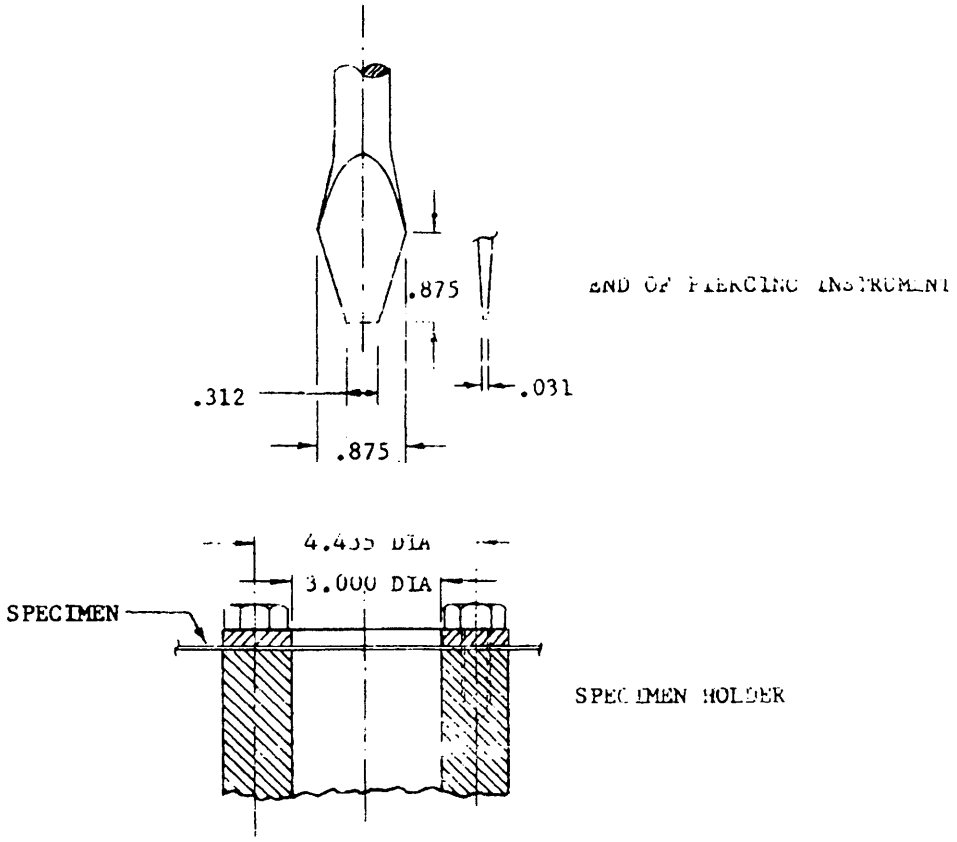


FIGURE 3 - PIERCING INSTRUMENT AND SPECIMEN HOLDER
(DIMENSIONS IN INCHES)

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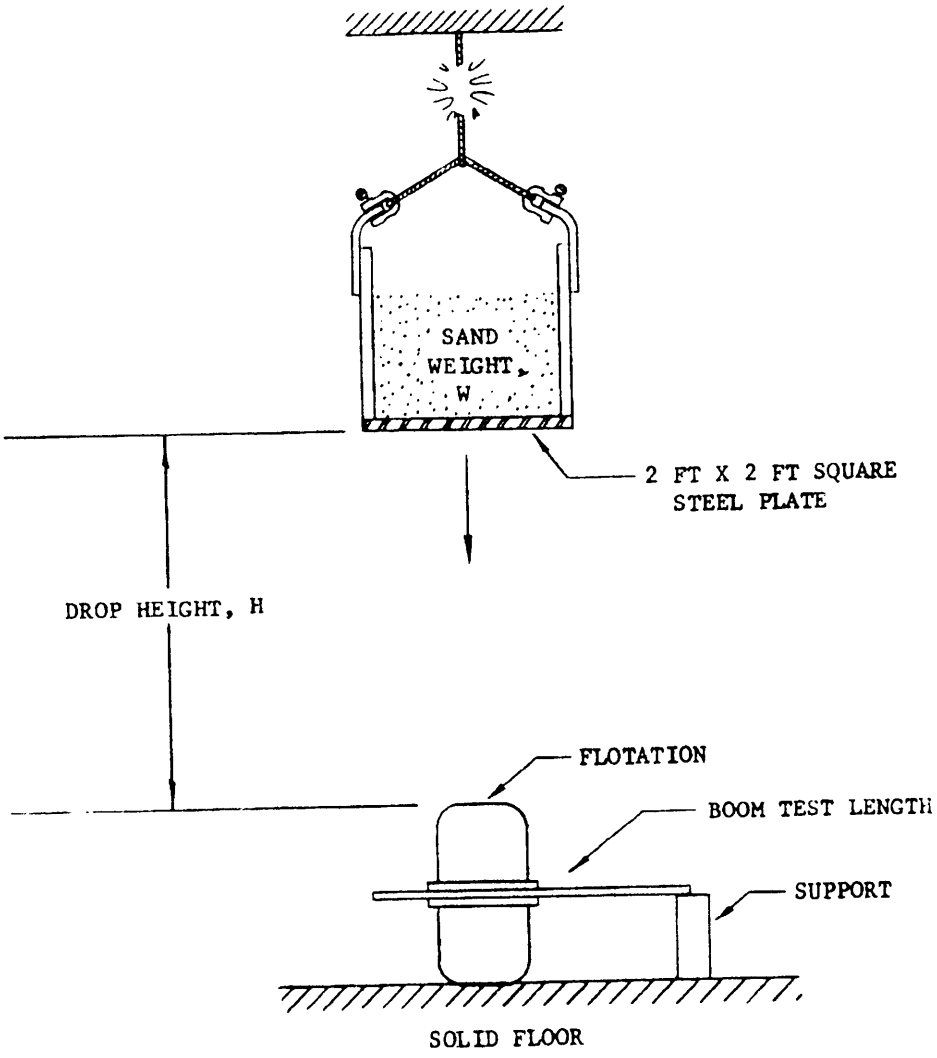
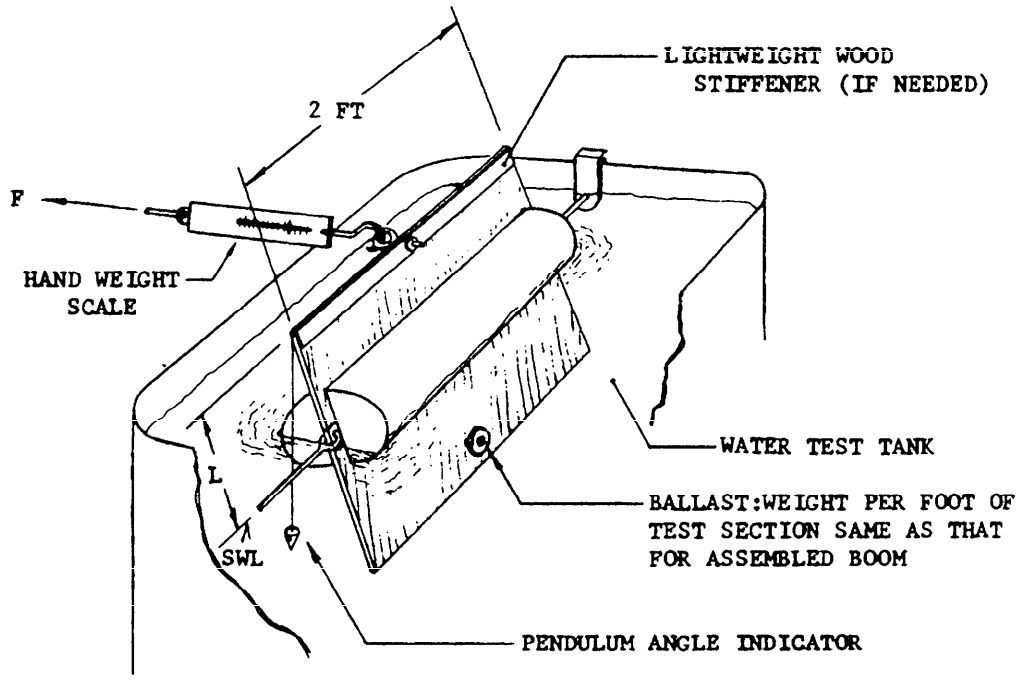


FIGURE 4 - BUMPERING TEST

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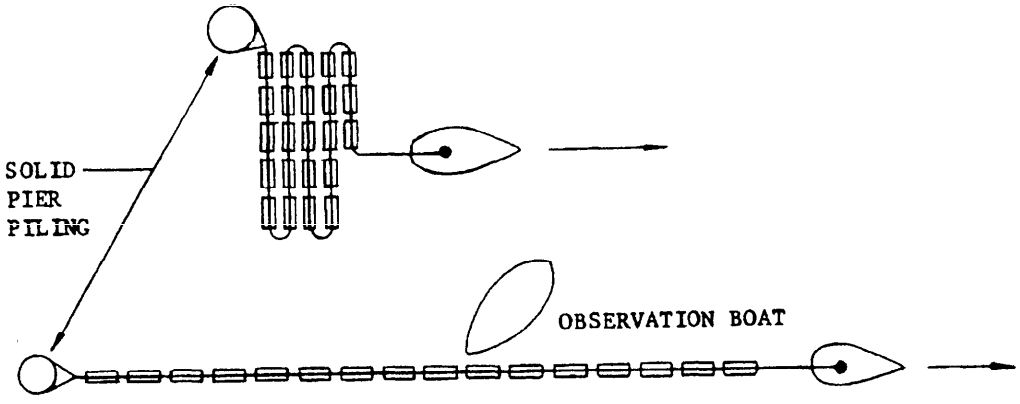


STATIC BOOM RIGHTING MOMENT, M, IS GIVEN BY:

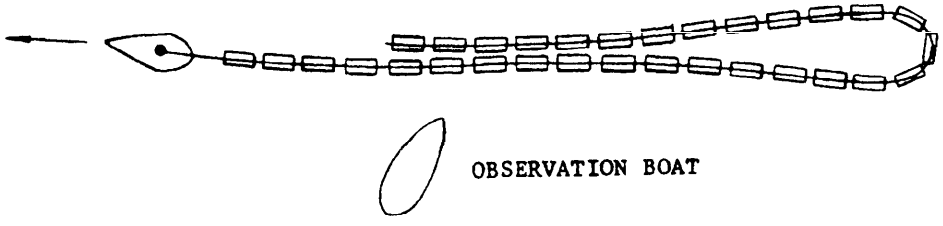
$$M(ft\text{-}lb) = L(ft) \times F(lb) \quad @ \ 30^{\circ} \text{ ROLL ANGLE}$$

FIGURE 5 - STATIC RIGHTING MOMENT TEST

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(a) TOW FROM RAFTED STOWAGE POSITION



(b) TOW IN 180° TURN

FIGURE 6 - TOWING MANEUVERS

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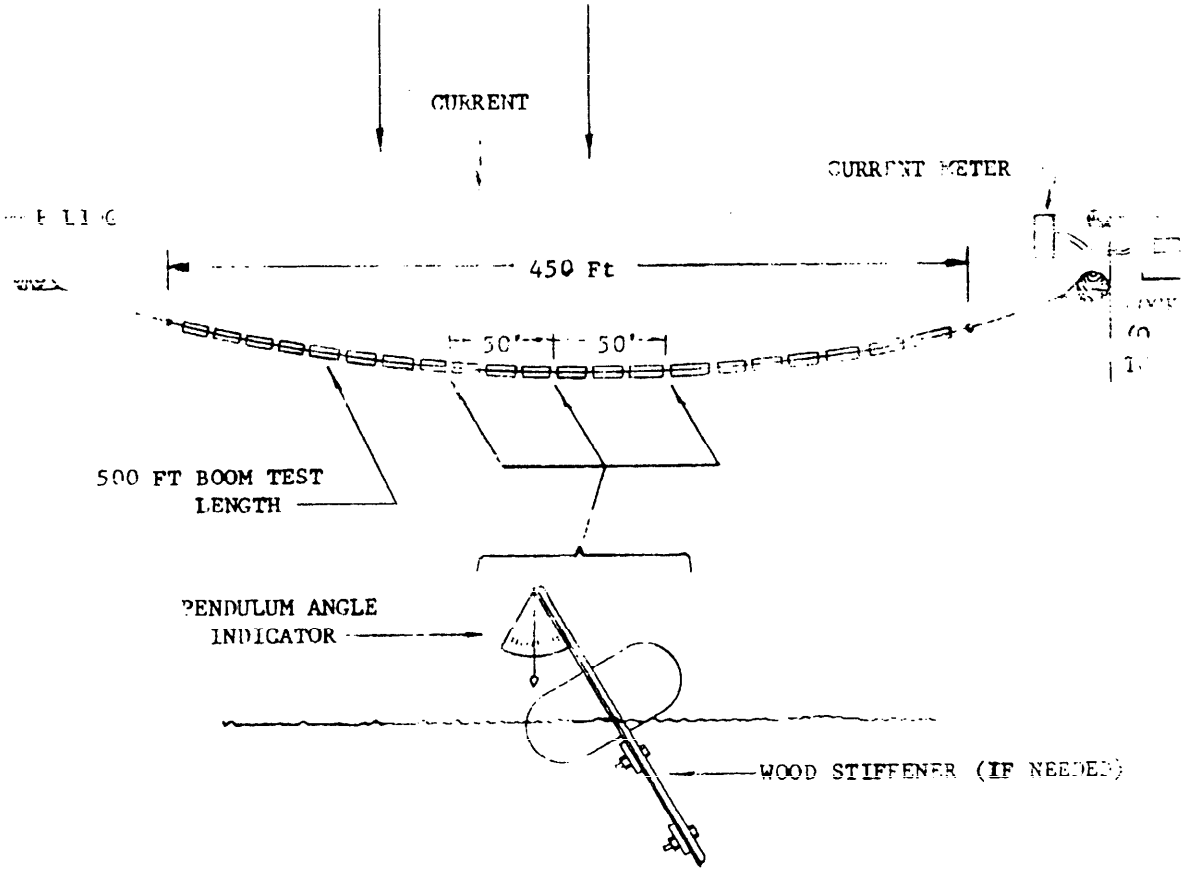


FIGURE 7 - ROLL STABILITY IN CURRENT

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